Labor Supply and Consumption under Uncertainty: Prudence Reconsidered*

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Abstract

This paper examines how variations in labor supply can be used to self insure against individual specific risks, and what implications this has for the magnitude of precautionary savings. It is found, maybe surprisingly, that the effects of precautionary saving can be significantly underestimated if wages fluctuate and labor supply is assumed fixed when it is not. This is the case even if utility is separable in consumption and leisure. The precautionary effects on leisure are similar to those on consumption. Agents work more today and less tomorrow if risk increases. Moreover, the standard definition of prudence is shown not to be applicable in the current setting. Consequently, a new definition of prudence is suggested.

Keywords: precautionary saving, prudence, risk aversion

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1. INTRODUCTION

How can and how do people use variations in labor supply to self insure against individual specific risks? In spite of the significant amount of work which has been devoted to both theoretical and quantitative studies of precautionary saving, this question seems to be unanswered.

Pratt (1964) showed that for an agent with utility function $u(c)$, where c is consumption, $-u_{cc}/u_c$ is a good measure of his absolute degree of risk aversion. While risk aversion is a concept closely related to an agent's utility, prudence and precautionary saving relate to the agent's behavior when facing uncertainty. Leland (1968) and Sandmo (1970) first formalized this concept and showed that a positive third derivative of the utility function is crucial for obtaining positive precautionary saving. Kimball (1990) paralleled Pratt's analysis and showed that a good measure of the absolute degree of prudence is $-u_{ccc}/u_{cc}$.

In this paper, I examine the precautionary behavior of agents who can choose their labor supply in each period. First, I look at the concepts of risk aversion and prudence in this setting, and relate them to Kimball's study. Then, the theoretical implications of prudence are analyzed in a simple two-period model. In particular, I examine how agents use variations in labor supply to self insure against risk, and what implications this has for the magnitude of precautionary savings.

The standard measure of prudence, as defined by Kimball, is not directly applicable in this setting. The reason is that it measures precautionary strength in terms of the uncertain variable. A natural requirement of the prudence measure is that it indicates how decision variables respond to uncertainty. For example, when studying precautionary saving and wage uncertainty, we want an agent with high prudence to save more than an agent with low prudence. The standard definition of prudence works well when labor supply is fixed, since wage and endowment shocks are then equivalent. But if labor supply is flexible, prudence needs to be measured in the same terms as savings and not in terms of the wage rate. In the next section I look in to this problem and suggest an alternative definition of prudence, which will fulfill the requirement under more general conditions.

Eaton and Rosen (1980) considered a framework where agents determine labor supply before wage uncertainty is resolved. They showed that labor supply can increase in response to increased wage uncertainty if risk aversion is sufficiently high. I show here that when allowing for an interaction between labor supply and saving decisions, increased wage uncertainty unambiguously lowers future labor supply. In an other related paper, Bodie, Merton, and Samuelson (1992) analyzed how labor supply flexibility influences investors' portfolio decisions. One finding was that greater labor supply flexibility induces more risk taking. Therefore, one might expect that greater labor supply flexibility also would make agents less prudent. However, this will not always be the case.

The analytical and numerical investigations in this paper show that the degree of prudence often will be underestimated if wages fluctuate and labor supply incorrectly is assumed to be fixed. Of course, flexibility is always good for the agents, so even if their precautionary savings increase, their expected utility is higher with flexible labor supply. With fixed labor supply, all effects of a negative shock must be absorbed by consumption. With flexible labor supply, hours worked can be adjusted to alleviate the effect of the shock. With the same

argument we note that a certain amount of saving is less costly for agents when labor supply is flexible. Therefore, agents with flexible labor supply are willing to expose themselves to more risk but they can more easily save to self insure against the uncertainty.

The usual measure of prudence is closely related to risk aversion. Kimball and Weil (1992) broke that link, and showed that both high risk aversion and high intertemporal elasticity of substitution tend to imply much prudence. The present paper illustrates this point. If an agent has decreasing absolute risk aversion, he can insure against wage fluctuations by bringing much wealth into the risky period, and a high intertemporal elasticity of substitution makes it less costly for the agent to shift wealth across periods.

Aiyagari (1994) found the precautionary savings effects to be modest in a quantitative general equilibrium model with fixed labor supply, at least for plausible parameterizations. As noted above, ignoring labor supply flexibility can make us underestimate the amount of precautionary saving. Moreover, this effect is particularly large when risk aversion is low. So, this paper indicates that Aiyagari's results must be reevaluated if we believe that labor supply is a decision variable for the individuals.

The remainder of the paper is structured as follows. In Section 2, I suggest a way of measuring risk and prudence in this framework. Thereafter, in Section 3, I use the measure of prudence to look at a two-period economy where there are endowment or wage shocks in the second period. This results in implications for the amount of precautionary savings for agents with flexible and fixed labor supply, respectively. Numerical examples show that agents with flexible labor supply often choose to hold more precautionary savings than agents who cannot choose their labor supply. In Section 4, I consider the effects uncertainty has on labor supply. It is shown that more wage uncertainty unambiguously has a positive effect on labor supply today but a negative effect on future labor supply. Section 5 concludes.

2. MEASURING RISK AVERSION AND PRUDENCE

This section starts by showing how to measure risk aversion in a framework that can allow for labor supply flexibility. This is a straightforward application of Pratt's (1964) analysis. Thereafter, an example demonstrates that the standard definition of prudence is not applicable in this setting. Finally, a new definition of prudence is suggested and contrasted to Kimball's (1990) definition.

Consider an agent who has an indirect utility function $V(a, \varepsilon)$, where a is wealth in the beginning of the period and ε is a mean zero shock during the period. In looking for a concept of risk aversion, we follow Pratt (1964) and define π as the premium the agent is prepared to pay to get $\varepsilon = 0$ with certainty, that is

$$V(a - \pi, 0) = \mathrm{E}_\varepsilon V(a, \varepsilon),$$

where E_ε denotes the expectation over ε. Let $o(x)$ denote terms that vanish faster than x as x approaches zero. Denote the variance of ε by σ^2, and expand V around a and $\varepsilon = 0$ to get

$$V(a - \pi, 0) = V(a, 0) + V_a(a, 0)(-\pi) + o\left(\sigma^2\right), \tag{1}$$

$$\begin{aligned} \mathrm{E}_\varepsilon V(a,\varepsilon) &= \mathrm{E}_\varepsilon\left[V(a,0)+V_\varepsilon(a,0)\,\varepsilon+\frac{1}{2}V_{\varepsilon\varepsilon}(a,0)\,\varepsilon^2+o\left(\varepsilon^2\right)\right] \\ &= V(a,0)+\frac{1}{2}V_{\varepsilon\varepsilon}(a,0)\,\sigma^2+o\left(\sigma^2\right), \end{aligned} \tag{2}$$

where subscripts denote derivatives. We neglect terms of order $o(\sigma^2)$ and use the equality between (1) and (2), to find

$$\pi(a)=\frac{\sigma^2}{2}r(a),$$

where the degree of absolute risk aversion is defined as

$$r(a)=-\frac{V_{\varepsilon\varepsilon}(a,0)}{V_a(a,0)}. \tag{3}$$

While risk aversion is related to the disutility agents get from uncertainty, prudence describes agents' behavior when facing risk. Therefore, a natural requirement of the measure of prudence is that it connects the decision variable of interest to uncertainty. In most applications, we are interested in how saving reacts to risk. Consequently, we look for a measure of prudence, η, so that we can write savings as a function of η and σ^2.

Let us solve for precautionary savings in a simple two period model to demonstrate that the measure of prudence proposed by Kimball does not generally fulfill this requirement. Assume that the agent has a time-separable utility function, abstract from discounting and assume zero interest rates. Further, assume that the first and second period utility functions are identical and that the agent has no initial financial wealth. Let s be savings and ε a mean zero, second period shock, and define $\bar{v}(s)$ and $v(s,\varepsilon)$ as the indirect first and second period utility functions respectively. The agent's indirect life-time utility is then defined as

$$V(s,\varepsilon)\equiv\bar{v}(s)+v(s,\varepsilon).$$

The agent chooses savings in the first period, s, to maximize his expected indirect utility $\mathrm{E}_\varepsilon V(s,\varepsilon)$. The first order condition is

$$\bar{v}_s(s)+\mathrm{E}_\varepsilon v_s(s,\varepsilon)=0. \tag{4}$$

Expand v_s and $\bar{v}_s$ around $\varepsilon=0$ and $s=0$, and neglect terms of $o(\sigma^2)$,

$$\begin{aligned} \mathrm{E}_\varepsilon v_s(s,\varepsilon) &= \mathrm{E}_\varepsilon\left[v_s(0,0)+v_{ss}(0,0)\,s+v_{s\varepsilon}(0,0)\,\varepsilon+\frac{1}{2}v_{s\varepsilon\varepsilon}(0,0)\,\varepsilon^2+v_{ss\varepsilon}(0,0)\,s\varepsilon\right] \\ &= v_s(0,0)+v_{ss}(0,0)\,s+\frac{1}{2}v_{s\varepsilon\varepsilon}(0,0)\,\sigma^2, \end{aligned}$$

and

$$\bar{v}_s(s)=\bar{v}_s(0)+\bar{v}_{ss}(0)\,s.$$

Assume now that resources on average are spread equally between periods so that $s=0$ solves (4) when $\sigma^2=0$. Then $v_s(0,0)=-\bar{v}_s(0)$, and $v_{ss}(0,0)=\bar{v}_{ss}(0)$. Hence, if (4) is fulfilled, we have that

$$0=\left[-v_s(0,0)+v_{ss}(0,0)\,s\right]+\left[v_s(0,0)+v_{ss}(0,0)\,s+\frac{1}{2}v_{s\varepsilon\varepsilon}(0,0)\,\sigma^2\right].$$

We get the implied amount of precautionary saving in response to a small uncertainty by rearranging,

$$s = \frac{\eta\sigma^2}{4}. \tag{5}$$

where we have defined the degree of absolute prudence

$$\eta \equiv -\frac{v_{s\varepsilon\varepsilon}(0,0)}{v_{ss}(0,0)}.$$

This shows that the amount of precautionary saving is determined by the convexity of marginal utility in ε ($v_{s\varepsilon\varepsilon}$) and by the response of marginal utility to changes in the decision variable (v_{ss}). Note that this expression deviates from Kimball's $\eta = -\frac{v_{s\varepsilon\varepsilon}}{v_{s\varepsilon}}$. However, we will see that these expressions are equivalent if labor supply is fixed or if ε is an endowment shock.

To highlight the difference between Kimball's (1990) analysis and the present, let us consider a more general framework, similar to Kimball's. Let θ denote a stochastic variable and denote its mean by $\bar{\theta}$ such that $\theta = \bar{\theta} + \tilde{\theta}$, where $\tilde{\theta}$ is the stochastic component with variance σ_θ^2. Further, let δ denote the decision variable, and let $V(\delta, \theta)$ be the agent's indirect utility function. Finally, let E_θ denote expectations over θ.

The agent chooses δ to solve

$$\max_\delta \mathrm{E}_\theta V(\delta, \theta).$$

The first order condition is then

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = 0.$$

Now, Kimball measures prudence by the premium, ψ, which fulfills

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = V_\delta\left(\delta, \bar{\theta} - \psi\right). \tag{6}$$

That is, if the agent gets $\bar{\theta} - \psi$ for sure, instead of the uncertain θ, his decision will be the same.[1] As the previous example indicates, this definition of prudence is not always relevant for the study of precautionary saving. In that example, θ is the wage rate, and δ is first period savings or consumption. We are then interested in the effects uncertainty has on the agent's decision. So, the precautionary premium we are interested in is the difference, $\hat{\psi}$, between the agent's choice under certainty and the choice under uncertainty,

$$\mathrm{E}_\theta V_\delta(\delta, \theta) = V_\delta\left(\delta - \hat{\psi}, \bar{\theta}\right). \tag{7}$$

Expanding these expressions around δ and $\bar{\theta}$ and neglecting terms of order $o(\sigma_\theta^2)$, we find

$$\hat{\psi}(\delta) = -\frac{V_{\delta\theta\theta}\left(\delta, \bar{\theta}\right)}{V_{\delta\delta}\left(\delta, \bar{\theta}\right)}\frac{\sigma_\theta^2}{2}.$$

[1] I only consider the "equivalent premium" here. An alternative is "compensating premia", which are additions to the uncertain variable.

Consequently, we define the degree of prudence as

$$\eta(\delta) \equiv -\frac{2V_{\delta\theta\theta}\left(\delta,\bar{\theta}\right)}{V_{\delta\delta}\left(\delta,\bar{\theta}\right)}.$$

In our previous example this amounts to

$$\eta(s) = -\frac{2V_{s\varepsilon\varepsilon}}{V_{ss}}.$$

Since $V_{ss} = 2v_{ss}$ and $V_{s\varepsilon\varepsilon} = v_{s\varepsilon\varepsilon}$ this can be rewritten as

$$\eta(s) = -\frac{v_{s\varepsilon\varepsilon}}{v_{ss}}.$$

The definition of prudence proposed in (7) is thus consistent with what we found in the example (cf. equation [5]), while Kimball's definition in (6) typically is not.

3. FLEXIBILITY AND PRECAUTIONARY BEHAVIOR

The first question we ask is how labor supply flexibility influences precautionary saving. The agent is assumed to have a time separable utility function, $u(c,l)$, where c is his consumption of goods, and l is his consumption of leisure. We restrict the study to utility functions which are strictly concave, i.e. $u_{cc}, u_{ll} < 0$, $u_{cc}u_{ll} - u_{cl}^2 > 0$, and 'precautionary', i.e. $u_c, u_l, u_{ccc}, u_{lll} > 0$. Further, the agent is assumed to have one unit of time to dispose of, and throughout the analysis we assume interior solutions for the leisure choice.

3.1. Endowment Uncertainty

Let us first consider the effects of endowment uncertainty on the agent's behavior. Assume that first and second period wage rates are equal and non-stochastic, $w_1 = w_2 = w$. The shock ε is now a second period endowment shock, so

$$c_1 = (1-l_1)w - s, \tag{8}$$
$$c_2 = (1-l_2)w + s + \varepsilon. \tag{9}$$

The agent chooses consumption and labor supply to maximize

$$u(c_1,l_1) + \mathrm{E}_\varepsilon u(c_2,l_2).$$

Given the time separable utility, the leisure choice in each period is a function of contemporaneous consumption and wages,

$$u_l(c_t,l_t) = w_t u_c(c_t,l_t). \tag{10}$$

This, together with (8), (9), and a fixed w, defines the leisure choices as functions of savings and the shock,

$$l_1 \equiv L_1(s),$$
$$l_2 \equiv L_2(s,\varepsilon).$$

The indirect first period utility function is then

$$\bar{v}(s) = u\left[(1 - L_1(s))\, w - s, L_1(s)\right],$$

and the indirect second period utility is

$$v(s, \varepsilon) = u\left[(1 - L_2(s, \varepsilon))\, w + s + \varepsilon, L_2(s, \varepsilon)\right].$$

Define

$$L_\varepsilon \equiv \frac{\partial L_2(s, \varepsilon)}{\partial \varepsilon}, \text{ and } L_s \equiv \frac{\partial L_2(s, \varepsilon)}{\partial s}.$$

Now, note that $L_\varepsilon = L_s$ and both s and ε enter additively to c_2. Hence $v_{s\varepsilon\varepsilon} = v_{sss}$. We use the Envelope theorem on the indirect second period utility function to get

$$v_s = u_c.$$

Let us restrict our analytical studies to utility functions which are additively separable in consumption and leisure so that $u_{cl} = 0$. We then get

$$\begin{aligned} v_{ss} &= (1 - L_\varepsilon w)\, u_{cc}, \\ v_{s\varepsilon\varepsilon} &= (1 - L_\varepsilon w)^2 u_{ccc} - L_{\varepsilon\varepsilon} w u_{cc}. \end{aligned}$$

Hence

$$\eta^{\text{flex}} = -\frac{v_{s\varepsilon\varepsilon}}{v_{ss}} = -\frac{(1 - L_\varepsilon w)\, u_{ccc}}{u_{cc}} + \frac{L_{\varepsilon\varepsilon} w}{1 - L_\varepsilon w}.$$

By differentiating (10) we can show that $0 < L_\varepsilon w < 1$, and

$$L_{\varepsilon\varepsilon} = L_\varepsilon (1 - L_\varepsilon w)^2 \frac{u_{ccc}}{u_{cc}} - \frac{L_\varepsilon^3 u_{lll}}{w u_{cc}}$$

so that

$$\eta^{\text{flex}} = -\frac{(1 - L_\varepsilon w)^2 u_{ccc}}{u_{cc}} - \frac{L_\varepsilon^3 u_{lll}}{(1 - L_\varepsilon w)\, u_{cc}} > 0.$$

If instead labor supply was fixed, $l_1 = l_2 = l$, we would of course get $L_\varepsilon = 0$ and $\eta^{\text{fix}} = -\frac{u_{ccc}}{u_{cc}}$. Hence, for additively separable preferences, there is a precautionary savings effect both when labor supply is flexible and when it is fixed. We also see that

$$\begin{aligned} \frac{\partial L_1}{\partial s} &= \frac{-w u_{cc}}{u_{ll} + w^2 u_{cc}} < 0, \\ \frac{\partial L_2}{\partial s} &= \frac{w u_{cc}}{u_{ll} + w^2 u_{cc}} > 0. \end{aligned}$$

That is, as risk increases, first period labor supply increases and average second period labor supply decreases. So there is a precautionary effect both on consumption goods and on leisure. The agent consumes less of both when future risk increases.

Here, flexible labor supply can affect the precautionary behavior through two mechanisms. When consumers decrease first period consumption, they also want to decrease first

period leisure because of the wealth effect. Moreover, second period labor supply is uncertain and therefore there is an additional precautionary effect when $u_{lll} > 0$. These effects will work towards more saving in the first period but this may, depending on the utility function, decrease the need for prudent consumption behavior. If there are no wealth effects (i.e. if $L_\varepsilon = 0$), and if uncertainty is of this form, then labor supply flexibility can be ignored.[2]

3.2. Wage Uncertainty

Consider now an economy where agents face an uncertain second period wage rate. The first period wage is $w_1 = w$ and the second period wage is $w_2 = w + \varepsilon$. Let us first assume that the agent can choose his labor input. Given the time separable utility, the leisure choice in each period is again a function of contemporaneous consumption and wages,

$$u_l(c_t, l_t) = w_t u_c(c_t, l_t). \tag{11}$$

Define first period savings as $s = (1 - l_1) w_1 - c_1$. Second period consumption is then $c_2 = (1 - l_2) w_2 + s$. The indirect first period utility function is

$$\bar{v}(s) = u[(1 - L_1(s)) w - s, L_1(s)],$$

and the indirect second period utility is

$$v(s, \varepsilon) = u[(1 - L_2(s, \varepsilon)) w_2 + s, L_2(s, \varepsilon)]. \tag{12}$$

To get further, we need a relationship between saving and leisure. Let us again assume that the utility function is separable in consumption and leisure so that $u_{cl} = 0$. The intra temporal equilibrium condition (11) holds for all ε and s, so

$$L_\varepsilon \equiv \frac{\partial L_2}{\partial \varepsilon} = \frac{(1 - L_2) w_2 u_{cc} + u_c}{u_{ll} + w_2^2 u_{cc}},$$

and

$$L_s \equiv \frac{\partial L_2}{\partial s} = \frac{w_2 u_{cc}}{u_{ll} + w_2^2 u_{cc}}. \tag{13}$$

From the Envelope theorem we have,

$$v_s(s, \varepsilon) = u_c.$$

After differentiating again we get

$$v_{ss}(s, \varepsilon) = (1 - L_s w_2) u_{cc} = \frac{u_{ll} u_{cc}}{u_{ll} + w_2^2 u_{cc}}. \tag{14}$$

and

$$v_{s\varepsilon}(s, \varepsilon) = (1 - L_2 - L_\varepsilon w_2) u_{cc}.$$

[2] There are no wealth effects if utility is $u(c, l) = V[c + H(l)]$,where V and H are increasing and concave functions. This corresponds to Example 3 in Bodie et. al. (1992).

A final differentiation with respect to ε yields

$$v_{s\varepsilon\varepsilon} = (1 - L_2 - L_\varepsilon w_2)^2 u_{ccc} - (2L_\varepsilon + L_{\varepsilon\varepsilon} w_2) u_{cc}$$

So

$$\eta^{\text{flex}} = \frac{-(1 - L - L_\varepsilon w)^2 u_{ccc}}{(1 - L_s w) u_{cc}} + \frac{2L_\varepsilon + L_{\varepsilon\varepsilon} w}{1 - L_s w}. \tag{15}$$

There are several mechanisms at work here. Labor supply changes due to a substitution effect, an income effect and a wealth effect. The substitution and income effects are due to fluctuations in the wage rate which changes the price on leisure relative to consumption. The wealth effect arises since prudent consumption behavior changes saving. Moreover, second period labor supply changes in response to wage shocks, so there will be a precautionary effect on leisure when $u_{ll} > 0$. Hence, first period leisure will tend to decrease and this tends to increase first period saving. However, for some utility functions, a decrease in leisure will tend to increase consumption of goods and hence decrease saving, so the total effect on saving, compared to the fixed labor supply scenario, is not clear.

If instead labor input is fixed at the certainty level from above, i.e. $l = L_1(0) = L_2(0,0)$, the indirect first period utility function is

$$\bar{v}(s) = u[(1 - l) w - s, l],$$

and the indirect second period utility is

$$v(s, \varepsilon) = u[(1 - l) w_2 + s, l].$$

Consequently,

$$v_s = u_c,$$

$$v_{ss} = u_{cc},$$

$$v_{s\varepsilon} = (1 - l) u_{cc},$$

and

$$v_{s\varepsilon\varepsilon} = (1 - l)^2 u_{ccc}.$$

Hence, for the fixed labor supply case we get

$$\eta^{\text{fix}} = -\frac{(1 - l)^2 u_{ccc}}{u_{cc}}. \tag{16}$$

In practice, the parameters in the utility function are often chosen so that either the intertemporal elasticity of substitution or the degree of risk aversion get a plausible value. These measures will typically not be the same when labor supply is flexible as when it is fixed. Therefore, it is not straightforward to compare utility functions under these different assumptions. If the utility function is additively separable in consumption and leisure, however, the parameter choices will likely be indifferent to the assumptions about labor supply flexibility. The additively separable utility is thus a natural starting point for the analysis.

ASSUMPTION 1: *Utility is separable in consumption and leisure,*

$$u(c,l) = q(c) + r(l),$$

where q and r are increasing and strictly concave functions.

If we also assume that the income, substitution, and wealth effects on leisure cancel, we can show that for small risks, precautionary saving is higher when labor supply is flexible than when it is fixed. This statement is formalized in Proposition 1.

PROPOSITION 1: *Assume that there is wage uncertainty, that the utility function fulfills Assumption 1, and that $L_\varepsilon = 0$. Then $\eta^{\text{flex}} > \eta^{\text{fix}}$.*

PROOF: This follows immediately from (15) and (16) since $0 < (1 - L_s w) < 1$ by (13). *Q.E.D.*

This Proposition applies to utility functions of the form $u(c,l) = \ln(c) + r(l)$, where r is a strictly concave function.

PROPOSITION 2: *If Assumption 1 is fulfilled and $q(c) = \ln(c)$ then $L_\varepsilon = 0$.*

PROOF: See the Appendix.

So far, I have compared savings decisions for agents who can choose labor supply to savings decisions for agents who are *restricted* not to vary labor input. The additively separable utility function allows us to also consider how changes in the intertemporal elasticity of leisure and labor input alters precautionary savings motives. Let us assume that the subutility for leisure belongs to the constant relative risk aversion class, and let $\bar{l}$ denote the leisure choice agents choose under certainty. Propositions 3 establishes that precautionary savings fall as leisure becomes less elastic in the utility function. Proposition 4 then shows that when leisure becomes totally inelastic in the utility function, agents choose the same savings as when labor supply is fixed by regulation. The proofs of these propositions are in the appendix.

ASSUMPTION 2: *Assume that*

$$\begin{aligned} q(c) &= \ln(c) \\ r(l) &= \frac{b\left(l^{1-\mu} - 1\right)}{1-\mu} \\ b &= \frac{\bar{l}^{1-\mu}}{1-\bar{l}}. \end{aligned}$$

PROPOSITION 3: *If Assumptions 1 and 2 are fulfilled, then*

$$\frac{\partial \eta}{\partial \mu} < 0.$$

PROPOSITION 4: *If Assumptions 1 and 2 are fulfilled, then*

$$\lim_{\mu \to \infty} \eta^{\text{flex}} = \eta^{\text{fix}}.$$

Consider now two agents, one with flexible labor supply and one with exogenously fixed labor supply. Both have chosen the same saving in the first period, and both have the same utility function, $u(c,l) = \left[\left(c^{\alpha} l^{1-\alpha}\right)^{1-\gamma} - 1\right] / (1-\gamma)$. Then both agents have the same degree of risk aversion for second period wage fluctuations.[3] Therefore, if the utility function is calibrated to match a certain degree of risk aversion, the calibration will be the same when labor supply is assumed to be endogenous as when labor supply is assumed to be fixed.

The unit elasticity of substitution class of utility functions is interesting since it allows us to relax the additive separability assumption and still show analytically that precautionary saving can be higher with flexible labor supply.

PROPOSITION 5: *Assume that* $u(c,l) = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-\gamma} - 1}{1-\gamma}$, $\gamma > 0$, *and that there is wage uncertainty. Further, assume that when labor supply is fixed, it is fixed at the level agents would choose under certainty. Then* $\eta^{\text{flex}} > \eta^{\text{fix}}$.

PROOF: See the Appendix.

In Figure 1, we see that, for small γ, the degree of prudence decreases as γ rises although calculations in the appendix show that the degree of risk aversion is increasing in γ.[4] This is in accordance with Kimball and Weil's (1992) finding that both high intertemporal elasticity of substitution and high risk aversion imply much precautionary saving. In the present example, the intertemporal elasticity of substitution for total expenditure goes to infinity as γ approaches zero, but the relevant degree of risk aversion does not approach zero. The agent's allocation of resources over time becomes less important when the intertemporal elasticity of substitution is high, and since the agent has decreasing absolute risk aversion (at least when $\gamma < 1$), he willingly saves much in the first period to minimize second period risk.

3.3. Examples

In order to quantify the precautionary effects and to get results for other utility functions than those considered previously, we have to rely on numerical solutions. I consider three general classes of utility functions,

$$u^{A}(c,l) = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-\gamma} - 1}{1-\gamma},$$

the constant elasticity of substitution class,

$$u^{B}(c,l) = \frac{\left[\alpha^{1-\nu} c^{\nu} + (1-\alpha)^{1-\nu} l^{\nu}\right]^{\frac{1-\gamma}{\nu}} - 1}{1-\gamma},$$

[3] This is shown in the appendix.

[4] It should, however, be noted that the relevance of the degree of prudence for precautionary saving is derived under the assumption that saving is small.

and the additively separable, constant relative risk aversion class,

$$u^C(c,l) = \frac{c^{1-\gamma}-1}{1-\gamma} + \frac{b\left(l^{1-\mu}-1\right)}{1-\mu},$$

where $0 < \alpha < 1$, $\gamma, \mu > 0$, and $\nu < 1$. In the CES utility function, $1/(1-\nu)$ is the elasticity of substitution between consumption and leisure. Recall that we have shown analytically that $s^{\text{flex}} > s^{\text{fix}}$ when there is wage uncertainty and the utility function belongs to class A.

Panels A, B, and C of Table I show results for different parameterizations of the respective utility functions when there are endowment shocks in the second period. The utility functions were calibrated with the same parameter values both when labor supply was flexible and when it was fixed. Table II shows the results for wage uncertainty. The variance of the endowment shocks is set to $\left(1-\bar{l}\right)^2$ times the variance of wage shocks so that the fixed labor supply experiments in Tables I and II are equivalent.

The tables show that allowing for labor flexibility can result in large changes in precautionary savings. For example, an agent with log utility ($\gamma = 1$ and utility class A) and flexible labor supply who is subject to wage uncertainty saves more than twice as much as an agent with fixed labor supply. The magnitudes of the precautionary effects in these experiments are small but so is the uncertainty agents face. The standard deviation of income is between 4 and 10 percent in these experiments.

Further, the tables indicate that labor supply flexibility lowers precautionary saving when there is endowment uncertainty while it increases saving when there is wage uncertainty. This has a natural interpretation. Because of the wealth effect, a negative endowment shock is offset by an increase in hours worked. But the substitution effect tends to decrease labor supply in response to negative wage shocks. Therefore, agents do not use variations in labor supply to self insure against wage risk to the same extent as they do against endowment risk.

As mentioned earlier, it is sometimes claimed that labor supply decisions can be neglected when utility is additively separable in consumption and leisure. We have seen that this claim is incorrect if we are looking at precautionary savings effects. However, consumption decisions are insensitive to labor supply decisions when utility is separable. For example, $c^{\text{flex}} = c^{\text{fix}}$ when $\gamma = 1$ and utility is of class A.

4. UNCERTAINTY AND LABOR SUPPLY

The previous analysis has indicated that more labor supply flexibility can increase the amount of precautionary savings. However, the exact ways in which agents use variations in labor supply to insure against shocks are diffuse since labor supply reacts to realized wage shocks. Let us therefore assume that second period labor supply must be chosen before uncertainty is resolved. Eaton and Rosen (1980) use the same timing, but they do not allow for precautionary saving. In contrast to Eaton and Rosen's analysis, we will see that increased wage uncertainty unambiguously lowers second period labor supply.

The second period wage is again uncertain, $w_2 = w + \varepsilon$, and we define savings as

$$s = (1-l_1)\,w_1 - c_1.$$

Consequently, agents maximize

$$u(c_1, l_1) + \mathrm{E}_\varepsilon u(c_2, l_2),$$

subject to

$$c_2 = (1 - l_2) w_2 + s.$$

We want to see how s and l_2 respond to increased uncertainty.[5] To do this, rewrite first period consumption and leisure as functions of savings, and let σ denote the uncertainty in w_2.[6] Then define

$$f(s) \equiv u_c[c_1(s), l_1(s)],$$

$$g(s, l_2; \sigma) \equiv \mathrm{E}_\varepsilon u_c(c_2, l_2),$$

and

$$h(s, l_2; \sigma) \equiv \mathrm{E}_\varepsilon [w_2 u_c(c_2, l_2) - u_l(c_2, l_2)].$$

The relevant first order conditions are then

$$f(s) = g(s, l_2; \sigma) \tag{17}$$

and

$$h(s, l_2; \sigma) = 0. \tag{18}$$

Proposition 6 states that agents choose more second period leisure and first period saving when wage uncertainty is increased.

PROPOSITION 6: *Assume that the utility function is additively separable in consumption and leisure. Then, for small σ,*

$$\frac{dl_2}{d\sigma} > 0,$$

and

$$\frac{ds}{d\sigma} > 0.$$

PROOF: See the Appendix.

By reducing second period labor supply, the agent avoids some risk, and for decreasing absolute risk aversion, increased saving also lowers risk. The agent can compensate the second period income loss (due to less work) by working more and consuming less in the first period.

[5] It is clear that both c_1 and l_1 decrease when savings increase.

[6] More precisely, σ is defined such that a higher σ implies more risk in the Rotschild - Stiglitz sence. So, if two stochastic variables have different σ, the variable with the highest σ can be constructed by adding a mean preserving spread to the other.

5. CONCLUDING REMARKS

This paper has considered precautionary behavior of agents with flexible labor supply in a simple two-period model. The main insight of the paper is that agents do not choose the same savings behavior when labor supply is a choice variable as when their labor supply is fixed. We have seen that ignoring labor supply flexibility can have significant effects in this framework. It remains for future research to evaluate these effects in general equilibrium and with long time horizons.

A potential problem with the comparisons conducted here is that risk may be measured differently under different assumptions about labor supply flexibility. If labor supply is believed to be fixed when it is not, uncertainty will be quantified as fluctuations in income rather than in wages. But income is then endogenous since it depends on work effort. However, this mismeasurement of risk will often reinforce the results found here since people work less in the second period when the wage increases (when $L_\varepsilon > 0$). In those cases, the idiosyncratic risk will be underestimated if labor supply incorrectly is assumed to be fixed.

Appendix A. Proofs and Calculations

A.1. Risk Aversion for Unit Elasticity of Substitution Utility

Assume that $u(c,l) = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-\gamma}-1}{1-\gamma}$, $\gamma > 0$, and that there is wage uncertainty. Let us first look at the case with flexible labor supply. We then get the equilibrium condition

$$l_2 = \frac{1-\alpha}{\alpha}\frac{c_2}{w_2}, \tag{A.1}$$

so

$$c_2 = \alpha\left(w_2 + s\right).$$

The indirect second period utility function is then

$$v(s, w_2) = \frac{c_2^{1-\gamma}}{1-\gamma} w_2^{-(1-\alpha)(1-\gamma)} K, \tag{A.2}$$

where K is a constant.

We use (A.2), to find the degree of absolute risk aversion for agents with flexible labor supply,[7]

$$r^{\text{flex}} = -\frac{v_{\varepsilon\varepsilon}}{v_s} = \frac{\alpha^2\gamma^{\text{flex}} + \alpha(1-\alpha)}{w}.$$

An agent with labor supply fixed at $l = 1-\alpha$ consumes

$$c_2 = \alpha w_2 + s,$$

[7] This is the risk attitude to what Drèze and Modigliani (1972) call "timeless" uncertainty, i.e. the risk agents face after having chosen first period saving. The expression is evaluated at $s = 0$.

and his degree of risk aversion is

$$r^{\text{fix}} = -\frac{(1-l)^2 u_{cc}}{u_c} = \frac{\alpha^2\gamma^{\text{fix}} + \alpha(1-\alpha)}{w}.$$

We see that the two agents have the same degree of risk aversion if they have the same γ.

A.2. Proof of Proposition 2

When $s = 0$ and $w_2 = w$, l and c are determined by the following two equations

$$\begin{aligned} l &= 1 - \frac{c}{w} \\ r'(l) &= \frac{w}{c} \end{aligned}$$

where the first equation is the budget constraint and the second equation is the first order condition for an optimum. It is immediately obvious that c/w must be constant when w changes. Hence $L_\varepsilon = 0$. *Q.E.D.*

A.3. Proof of Proposition 3

$L_\varepsilon = 0$ by Proposition 2. Using this in (15) we get

$$\eta = \frac{-(1-l)^2 u_{ccc}}{(1-L_s w) u_{cc}} = \frac{K_1}{1-L_s w} \tag{A.3}$$

where

$$K_1 \equiv \frac{(1-l)^2}{c} > 0.$$

In these equations, only L_s depends on γ. Hence

$$\frac{\partial \eta}{\partial \gamma} = \frac{wK_1}{(1-L_s w)^2}\frac{\partial L_s}{\partial \gamma}$$

and it is clear that

$$\text{sign}\left(\frac{\partial \eta}{\partial \gamma}\right) = \text{sign}\left(\frac{\partial L_s}{\partial \gamma}\right).$$

The derivative of second period leisure with respect to saving is given by (13),

$$L_s = \frac{w u_{cc}}{u_{ll} + w^2 u_{cc}}.$$

Consequently

$$\begin{aligned} \frac{\partial L_s}{\partial \gamma} &= -\frac{w u_{cc}}{(u_{ll} + w^2 u_{cc})^2}\frac{\partial u_{ll}}{\partial \gamma} \\ &= K_2 \frac{\partial(-\gamma B l^{-\gamma-1})}{\partial \gamma} \end{aligned}$$

where

$$K_2 \equiv -\frac{wu_{cc}}{\left(u_{ll} + w^2 u_{cc}\right)^2} > 0.$$

The coefficient B is defined as

$$B = \frac{l^\gamma}{1-l}$$

since $\bar{l} = l|_{s=0,w_2=w}$, so

$$\frac{\partial\left(-\gamma B l^{-\gamma-1}\right)}{\partial\gamma} = \frac{\partial\left[-\gamma l^{-1}\left(1-l\right)^{-1}\right]}{\partial\gamma} = -\frac{1}{l\left(1-l\right)} < 0.$$

This shows that

$$\frac{\partial\eta}{\partial\gamma} < 0.$$

Q.E.D.

A.4. Proof of Proposition 4

First note that η^{fix} is invariant to γ. This is seen from equation (16) which here yields

$$\eta^{\text{fix}} = \frac{\left(1-l\right)^2}{c}.$$

From (A.3) we get

$$\eta^{\text{flex}} = \frac{K_1}{1 - L_s\left(\gamma\right)w},$$

and (13) gives

$$L_s = \frac{w/c^2}{\gamma l^{-\gamma-1} + w^2/c^2}.$$

Since $0 < l < 1$, we see that

$$\lim_{\gamma\to\infty} \gamma l^{-\gamma-1} = \infty$$

and consequently that

$$\lim_{\gamma\to\infty} L_s = 0.$$

This verifies the Proposition,

$$\lim_{\gamma\to\infty} \eta^{\text{flex}} = K_1 = \frac{\left(1-l\right)^2}{c} = \eta^{\text{fix}}.$$

Q.E.D.

A.5. Proof of Proposition 5

We evaluate η^{flex} and η^{fix} and show that $\eta^{\text{flex}} > \eta^{\text{fix}}$. Differentiate (A.2) to get

$$\eta^{\text{flex}} = -\frac{v_{s\varepsilon\varepsilon}(0,0)}{v_{ss}(0,0)} = \frac{[1-\alpha(1-\gamma)][2-\alpha(1-\gamma)]}{\gamma w}.$$

When labor supply is fixed,

$$-\frac{u_{ccc}}{u_{cc}} = \frac{2-\alpha(1-\gamma)}{\mathrm{E}_\varepsilon c_2} = \frac{2-\alpha(1-\gamma)}{\alpha w}.$$

So, from (16)

$$\eta^{\text{fix}} = \frac{\alpha[2-\alpha(1-\gamma)]}{w}.$$

We know that $1-\alpha > 0$ and $\gamma > 0$ so

$$\frac{1-\alpha(1-\gamma)}{\gamma} > \alpha.$$

Hence $\eta^{\text{flex}} > \eta^{\text{fix}}$. *Q.E.D.*

A.6. Proof of Proposition 6

Totally differentiate (17) and (18) to get

$$\frac{dl_2}{d\sigma} = -\frac{g_\sigma h_s + h_\sigma(f_s - g_s)}{(f_s - g_s)h_l + g_l h_s}, \tag{A.4}$$

and

$$\frac{ds}{d\sigma} = \frac{g_\sigma h_l - g_l h_\sigma}{(f_s - g_s)h_l + g_l h_s}. \tag{A.5}$$

We know that $f_s > 0$, $g_s < 0$, $g_l > 0$, $h_s < 0$, and $h_l > 0$. Since we have assumed that $u_{ccc} > 0$, we see that $u_c(c_2, l_2)$ is convex in w_2, and hence that

$$g_\sigma > 0,$$

where g_σ is the derivative of g with respect to its third argument. Uncertainty has unclear effects on h, however. If

$$(1-l_2)[w_2(1-l_2)u_{ccc} + 2u_{cc} - (1-l_2)u_{lcc}]$$

is positive, then $w_2 u_c - u_l$ is convex in w_2, and if the expression is negative, $w_2 u_c - u_l$ is concave.

For additively separable utility and small s and σ, the denominator in (A.4) and (A.5) is positive since then

$$\begin{aligned} f_s &= -g_s \\ g_l &= -w g_s \\ h_s &= -g_l \\ h_l &= w g_l - u_{ll}(l_2), \end{aligned}$$

that is,

$$(f_s - g_s)\, h_l + g_l h_s = 2g_s \left(w^2 g_s + u_{ll}\right) - w^2 g_s^2 = w^2 g_s^2 + 2g_s u_{ll} > 0.$$

It remains to show that $g_\sigma h_l - g_l h_\sigma > 0$. Expanding g and h around w, we get for small σ

$$g_\sigma = \chi_1 u_{ccc}$$

and

$$h_\sigma = \chi_1 w u_{ccc} + \chi_2 u_{cc}$$

where $\chi_1 > 0$ and $\chi_2 > 0$. Hence

$$\begin{aligned} g_\sigma h_l - g_l h_\sigma &= \chi_1 u_{ccc} \left(w g_l - u_{ll}\right) - g_l \left(\chi_1 w u_{ccc} + \chi_2 u_{cc}\right) \\ &= -\chi_1 u_{ccc} u_{ll} - g_l \chi_2 u_{cc} > 0. \end{aligned}$$

This shows that $\frac{ds}{d\sigma} > 0$. It then follows immediately that also $\frac{dl_2}{d\sigma} > 0$. *Q.E.D.*

Appendix B. Notes on Calculations

[*This appendix is to be removed*]

B.1. Notes on Section 3.1

Second period consumption is given by the budget constraint,

$$c_2 = \left[1 - L_2\left(s, \varepsilon\right)\right] w + s + \varepsilon.$$

So,

$$\begin{aligned} \frac{\partial c_2}{\partial s} &= 1 - L_s w, \\ \frac{\partial c_2}{\partial \varepsilon} &= 1 - L_\varepsilon w. \end{aligned}$$

Equation (10) holds for all s and ε so we can differentiate with respect to s and ε to get (assuming $u_{cl} = 0$),

$$\begin{aligned} u_{ll} L_s &= \left(1 - L_s w\right) w u_{cc}, \\ u_{ll} L_\varepsilon &= \left(1 - L_\varepsilon w\right) w u_{cc}, \end{aligned}$$

which implies that

$$L_s = L_\varepsilon = \frac{w u_{cc}}{u_{ll} + w^2 u_{cc}}. \tag{B.6}$$

This also shows that $0 < w L_\varepsilon < 1$ since

$$w L_\varepsilon = \frac{w^2 u_{cc}}{u_{ll} + w^2 u_{cc}},$$

and both u_{cc} and u_{ll} are assumed to be strictly negative.

Differentiate (B.6) to get

$$\begin{aligned}
L_{\varepsilon\varepsilon} &= \frac{w\left(1-L_{\varepsilon}w\right)u_{ccc}}{u_{ll}+w^{2}u_{cc}}-\frac{wL_{\varepsilon}u_{lll}u_{cc}+w^{3}\left(1-L_{\varepsilon}\right)u_{ccc}u_{cc}}{\left[u_{ll}+w^{2}u_{cc}\right]^{2}} \\
&= \frac{w\left(1-L_{\varepsilon}w\right)u_{ccc}\left(u_{ll}+w^{2}u_{cc}\right)-wL_{\varepsilon}u_{lll}u_{cc}-w^{3}\left(1-L_{\varepsilon}\right)u_{ccc}u_{cc}}{\left[u_{ll}+w^{2}u_{cc}\right]^{2}} \\
&= \frac{w\left(1-L_{\varepsilon}w\right)u_{ccc}u_{ll}-wL_{\varepsilon}u_{lll}u_{cc}}{\left[u_{ll}+w^{2}u_{cc}\right]^{2}} \\
&= \left\{\text{use } u_{ll}=\left(1-L_{\varepsilon}w\right)wu_{cc}/L_{\varepsilon}\right\} \\
&= \frac{w^{2}\left(1-L_{\varepsilon}w\right)^{2}u_{ccc}u_{cc}-wL_{\varepsilon}^{2}u_{lll}u_{cc}}{L_{\varepsilon}\left[\left(1-L_{\varepsilon}w\right)wu_{cc}/L_{\varepsilon}+w^{2}u_{cc}\right]^{2}} \\
&= \frac{w^{2}L_{\varepsilon}\left(1-L_{\varepsilon}w\right)^{2}u_{ccc}u_{cc}-wL_{\varepsilon}^{3}u_{lll}u_{cc}}{\left(wu_{cc}\right)^{2}} \\
&= \frac{L_{\varepsilon}\left(1-L_{\varepsilon}w\right)^{2}u_{ccc}}{u_{cc}}-\frac{L_{\varepsilon}^{3}u_{lll}}{wu_{cc}}.
\end{aligned}$$

B.2. Notes on Section 3.2

Now, second period consumption is

$$c_2 = \left[1-L_2\left(s,\varepsilon\right)\right]w_2+s,$$

where

$$w_2 = w+\varepsilon.$$

So,

$$\begin{aligned}
\frac{\partial c_2}{\partial s} &= 1-L_s w_2, \\
\frac{\partial c_2}{\partial \varepsilon} &= 1-L-L_{\varepsilon}w_2.
\end{aligned}$$

The calculations in the section are then straightforward.

B.3. Notes on Appendix A.3.

Let

$$H \equiv 1-l_2.$$

Expand

$$g\left(s,l_2;\sigma\right) = \left[\mathrm{E}_{\varepsilon}u_c\left(c_2,l_2\right)\right]$$

and

$$h\left(s,l_2;\sigma\right) \equiv \mathrm{E}_{\varepsilon}\left[w_2u_c\left(c_2,l_2\right)-u_l\left(c_2,l_2\right)\right]$$

around w for small s and σ to get

$$g = u_c + \frac{1}{2}\sigma_\varepsilon^2 H^2 u_{ccc}$$

and

$$h = wu_c - u_l + \frac{1}{2}\sigma_\varepsilon^2 \left(wH^2 u_{ccc} + 2Hu_{cc}\right).$$

Define

$$\chi_1 \equiv \sigma_\varepsilon H^2$$

and

$$\chi_2 \equiv 2\sigma_\varepsilon H.$$

We then get

$$g_\sigma = \chi_1 u_{ccc}$$

and

$$h_\sigma = \chi_1 w u_{ccc} + \chi_2 u_{cc}.$$

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TABLE I

PRECAUTIONARY EFFECTS WITH ENDOWMENT UNCERTAINTY

	Parameters		s^{flex}	s^{fix}	c_1^{flex}	c_1^{fix}	$\bar{c}_2^{\text{flex}}$	$\bar{c}_2^{\text{fix}}$	l_1	$\bar{l}_2$
A		γ								
		0.5	.006	.016	3.997	3.984	4.002	4.020	.600	.600
		1.0	.008	.020	3.997	3.980	4.003	4.021	.599	.600
		2.0	.012	.025	3.995	3.975	4.005	4.024	.599	.601
		5.0	.025	.037	3.990	3.963	4.009	4.037	.598	.601
		10.0	.046	.057	3.981	3.943	4.017	4.058	.597	.602
B	ν	γ								
	0.0	1.0	.008	.020	3.997	3.980	4.003	4.020	.599	.600
	−2.0	1.0	.006	.026	3.997	3.974	4.004	4.030	.600	.601
	0.5	1.0	.009	.017	3.996	3.982	4.003	4.015	.599	.600
	0.0	0.5	.007	.019	3.997	3.980	4.002	4.017	.600	.600
	0.0	2.0	.012	.024	3.995	3.976	4.005	4.025	.599	.601
	−2.0	2.0	.011	.034	3.995	3.966	4.005	4.036	.599	.601
C	μ	γ								
	1.0	1.0	.008	.020	3.997	3.980	4.003	4.020	.599	.600
	5.0	1.0	.012	.020	3.991	3.980	4.014	4.020	.600	.600
	2.0	2.0	.011	.030	3.995	3.970	4.005	4.031	.599	.601
	5.0	2.0	.018	.030	3.989	3.970	4.012	4.031	.599	.600
	5.0	5.0	.023	.061	3.991	3.939	4.010	4.064	.599	.601

Note: $w = 10$, $\varepsilon \sim N\left[0, \left(1-\bar{l}\right)^2\right]$, and $\alpha = 0.40$ for utility function A while α depends on ν for utility function B such that consumption is 4 and leisure is .6 under certainty. s = first period saving, superscripts refer to the model with flexible and fixed labor supply respectively.

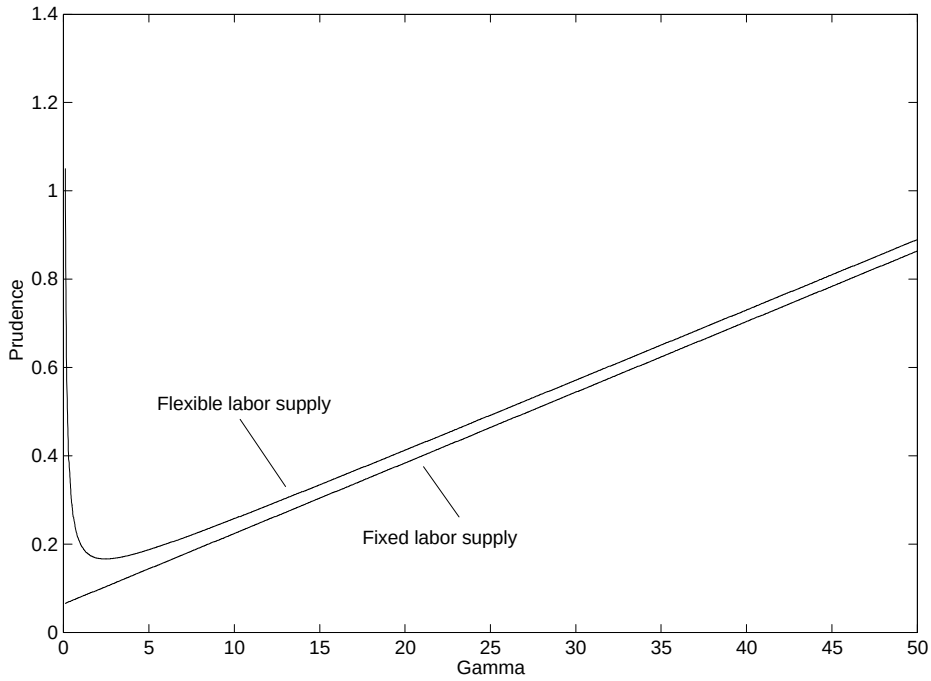
TABLE II

PRECAUTIONARY EFFECTS WITH WAGE UNCERTAINTY

	Parameters		s^{flex}	s^{fix}	c_1^{flex}	c_1^{fix}	$\bar{c}_2^{\text{flex}}$	$\bar{c}_2^{\text{fix}}$	l_1	$\bar{l}_2$
A		γ								
		0.5	.068	.017	3.973	3.983	4.030	4.019	.596	.604
		1.0	.053	.021	3.979	3.979	4.020	4.020	.597	.603
		2.0	.043	.025	3.983	3.975	4.017	4.024	.597	.603
		5.0	.044	.034	3.982	3.966	4.023	4.039	.597	.603
		10.0	.068	.059	3.973	3.941	4.025	4.057	.596	.604
B	ν	γ								
	0.0	1.0	.047	.019	3.981	3.981	4.018	4.018	.597	.603
	−2.0	1.0	.048	.026	3.981	3.974	4.012	4.024	.597	.602
	0.5	1.0	.048	.016	3.981	3.984	4.030	4.014	.597	.604
	0.0	0.5	.066	.016	3.974	3.984	4.027	4.017	.596	.604
	0.0	2.0	.041	.023	3.984	3.977	4.015	4.021	.598	.602
	−2.0	2.0	.041	.032	3.983	3.968	4.010	4.032	.597	.602
C	μ	γ								
	1.0	1.0	.047	.019	3.981	3.981	4.018	4.018	.597	.603
	5.0	1.0	.024	.019	3.981	3.981	4.019	4.019	.599	.601
	2.0	2.0	.040	.028	3.984	3.973	4.011	4.028	.598	.602
	5.0	2.0	.032	.028	3.980	3.972	4.017	4.028	.599	.601
	5.0	5.0	.046	.054	3.982	3.946	4.013	4.058	.597	.602

Note: $w = 10$, $\varepsilon \sim N(0,1)$, and $\alpha = 0.40$ for utility function A while α depends on ν for utility function B, and b depends on γ and μ for utility function C, such that consumption is 4 and leisure is .6 under certainty. s = first period saving, superscripts refer to the model with flexible and fixed labor supply respectively.

Figure 1
Degree of prudence with wage uncertainty and unit elastic utility



The graph relates the degrees of prudence for flexible and fixed labor supply (η^{flex} and η^{fix}) to the parameter γ in the utility function

$$u(c,l) = \frac{\left(c^{\alpha} l^{1-\alpha}\right)^{1-\gamma} - 1}{1-\gamma}$$